Solar N appendix

Sun · N · downloadable appendix

Document	Solar N appendix
Subject	Solar photospheric abundance appendix · N
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Date	2026-10-29
Publisher	The Exoplanet Codex · exoplanetcodex.org
Version	feed v1.80
Source	science commit 46fd923d5101
Web	https://exoplanetcodex.org/systems/sol/elements/n/

Solar nitrogen: 7.855 ± 0.049

Asplund, Amarsi & Grevesse 2021: 7.83 ± 0.07 · this measurement +0.025 dex

The solar N I lines are weak (0.3–1% deep) and carry CN absorption inside their own profiles, so the result depends on molecular opacity and on continuum placement at the 0.1% level. Both are now modelled: CN is synthesised with the line, and the continuum is placed locally around every line. For comparison, Asplund et al. 2021 give 7.77 from N I after empirically rescaling the CN blends, and Lodders, Bergemann & Palme 2025 adopt 7.94 ± 0.11 from the two least-blended lines. The combined nitrogen headline follows Asplund et al. 2021 Sect. 2.3: the mean of the atomic and molecular results. Its uncertainty is the larger of half their separation and the two product errors combined, so the combined bar is never smaller than the individual budgets (CN blending, continuum, telluric) support.

Combined the way Asplund, Amarsi & Grevesse (2021) combine it: unweighted mean of atomic N I and the weighted molecular mean (NH, CN), both on the 3D scale; uncertainty half their range or their propagated error, whichever is larger (Asplund+2021 Sect. 2.3); 1D results moved to the 3D scale by Asplund+2021 Table 3 (3D minus 1D MARCS for the same indicator). Each indicator is represented by our best analysis of it.

N I: 7.904 ± 0.033 · IAG FTS · red-optical · N I 8629 + 8683 Å, the two least-blended lines (Magg et al. 2022, adopted by Lodders et al. 2025) · 3D NLTE (Amarsi)

CN: 7.806 ± 0.092 (1D 7.866 -0.060 to 3D) · Kitt Peak (1984, molecfit) · NIR · CN A–X (0–0) band, near-infrared (molecular band) · 1D LTE

Not measurable in our spectra: NH.

Highlighted Products

red-optical

7.904 ± 0.033

N I 8629 + 8683 Å, the two least-blended lines (Magg et al. 2022, adopted by Lodders et al. 2025) · 3D NLTE (Amarsi) · IAG FTS · Reference Grade

NIR

7.866 ± 0.092

CN A–X (0–0) band, near-infrared (molecular band) · 1D LTE · Kitt Peak (1984, molecfit) · Reference Grade

Line grades

Every product is measured on one of three grades of spectral lines and the grades are never mixed in one number.

Grade	What it means	This element
Reference Grade	lines from a published solar line set (Asplund et al. 2021 and similar), measured with the oscillator strengths the authors used	11 products -- CN A–X (0–0) band, near-infrared (molecular band); N I 8629 + 8683 Å, the two least-blended lines (Magg et al. 2022, adopted by Lodders et al. 2025)
Codex Grade	lines chosen by the Codex whose oscillator strength carries a published uncertainty (laboratory measurement, NIST accuracy class A-C, or the paper's own stated error), between 5% and 60% deep	1 product -- CN red system, visible (molecular band; diagnostic only)
Deep Grade	the same oscillator-strength rule on lines deeper than 60% -- stronger lines, more sensitive to damping and the model atmosphere	none for the Sun: every line in this grade is excluded as saturated or an outlier (see Excluded lines)

Error-bar Forest Plot

VIS — HARPS
A(N) dex; axis 3.90–10.69

CN red system, visible (molecular band; diagnostic only) · 1D LTE
Codex Grade · 194 lines



7.296
±3.363

VIS axis: 3.90 · 5.60 · 7.30 · 8.99 · 10.69 A(N) dex

red-optical — IAG FTS
A(N) dex; axis 7.59–8.24

N I 8629 + 8683 Å, the two least-blended lines (Magg et al. 2022, adopted by Lodders et al. 2025) · 1D LTE
Reference Grade · 2 lines



7.949
±0.030

N I 8629 + 8683 Å, the two least-blended lines (Magg et al. 2022, adopted by Lodders et al. 2025) · 3D NLTE (Amarsi)
Reference Grade · 2 lines



7.904
±0.033

N I 8629 + 8683 Å, the two least-blended lines (Magg et al. 2022, adopted by Lodders et al. 2025) · 1D NLTE (Bergemann)
Reference Grade · 1 line



7.935
±0.280

red-optical — Kitt Peak (Kurucz 2005)
A(N) dex; axis 7.59–8.24

N I 8629 + 8683 Å, the two least-blended lines (Magg et al. 2022, adopted by Lodders et al. 2025) · 1D LTE
Reference Grade · 2 lines



7.939
±0.034

N I 8629 + 8683 Å, the two least-blended lines (Magg et al. 2022, adopted by Lodders et al. 2025) · 3D NLTE (Amarsi)
Reference Grade · 2 lines



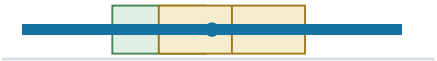
7.894
±0.034

N I 8629 + 8683 Å, the two least-blended lines (Magg et al. 2022, adopted by Lodders et al. 2025) · 1D NLTE (Bergemann)
Reference Grade · 1 line



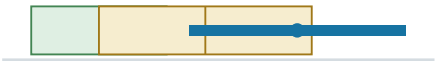
7.912
±0.244

red-optical — Kitt Peak (1984, molecfit)
A(N) dex; axis 7.59–8.24

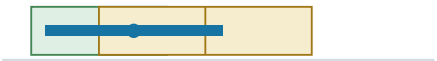
N I 8629 + 8683 Å, the two least-blended lines (Magg et al. 2022, adopted by Lodders et al. 2025) · 1D LTE Reference Grade · 2 lines		7.942 ±0.042
N I 8629 + 8683 Å, the two least-blended lines (Magg et al. 2022, adopted by Lodders et al. 2025) · 3D NLTE (Amarsi) Reference Grade · 2 lines		7.897 ±0.041
N I 8629 + 8683 Å, the two least-blended lines (Magg et al. 2022, adopted by Lodders et al. 2025) · 1D NLTE (Bergemann) Reference Grade · 1 line		7.910 ±0.286

red-optical axis: 7.59 · 7.76 · 7.92 · 8.08 · 8.24 A(N) dex

NIR — CRIRES+ J
A(N) dex; axis 7.73–8.18

CN A–X (0–0) band, near-infrared (molecular band) · 1D LTE Reference Grade · 185 lines		8.035 ±0.112

NIR — Kitt Peak (1984, molecfit)
A(N) dex; axis 7.73–8.18

CN A–X (0–0) band, near-infrared (molecular band) · 1D LTE Reference Grade · 215 lines		7.866 ±0.092

NIR axis: 7.73 · 7.84 · 7.95 · 8.07 · 8.18 A(N) dex

How to read this forest. Each row is one independent product, preserving its band, holding, engine, indicator and line set. Engines and indicator families are never averaged. Each bar is the uncertainty as Asplund et al. 2021 quote it: line-to-line scatter, the 3D-model term and the NLTE term (at least 0.03 dex). For a molecule the NLTE term is replaced by its dependence on the other element, and the 3D term comes from the 3D-to-1D difference Amarsi et al. 2021 publish for the same band. Atomic-data and observational terms are not in the bar, as in Asplund; they stay in the full error budget attached to the row. Grades describe the supplied line-pool provenance, not engine quality; evaluated theoretical CNO oscillator strengths are not relabelled as laboratory evidence. Quarantined, failed and unconstrained fits have no abundance mark. Experimental products remain explicitly not adopted. Literature references provide context, never tuning targets.

How this was measured

Every nitrogen value on this page comes from fitting synthetic spectra to the observed solar spectrum, line by line or band by band, with the nitrogen abundance as the free parameter. The synthesis includes the atomic lines and the molecular bands (CN, CH, C_2, OH and others) in every window, so blended absorption is modelled rather than attributed to the line being measured. The continuum is placed locally around every line by comparing the observation with the synthesis on its own highest pixels, and ground-based spectra are telluric-corrected across their full range.

Each indicator is measured independently on each solar spectrum and reported separately; the combined headline averages the atomic and molecular results, while individual products remain separate. Where a 3D or NLTE correction is published for the lines, it is applied as its own product (1D-LTE, 1D-NLTE and 3D-NLTE are separate rows). Every value carries a full uncertainty budget: fit, line strengths, microturbulence, continuum, fit

window, model atmosphere, telluric residual, blends and coupling to the other C/N/O abundances, and the spread between spectra, each measured by refitting with that one input changed. The nitrogen headline follows Asplund et al. 2021 Sect. 2.3: the atomic–molecular mean, with an uncertainty that is the larger of half their separation and the two product errors combined.

Band	What was measured	Solar spectra	Treatments	Lines	Result A(N)	Largest uncertainty terms (dex)
VIS	CN red system, visible (molecular band; diagnostic only)	HARPS	1D-LTE	194	7.296 (1D-LTE)	fit / line-to-line scatter 3.34, continuum placement 1.34
red-optical	N I 8629 + 8683 Å, the two least-blended lines (Magg et al. 2022, adopted by Lodders et al. 2025)	IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)	1D-LTE, 1D-NLTE, 3D-NLTE	1–2	7.894–7.904 (3D-NLTE)	continuum placement 0.18, microturbulence 0.03
NIR	CN A–X (0–0) band, near-infrared (molecular band)	CRIRES+ J, Kitt Peak (1984, molecfit)	1D-LTE	185–215	7.866–8.035 (1D-LTE)	line strengths (gf) 0.12, spread between spectra 0.09

Solar nitrogen context

Nitrogen is traced by atomic N I in the red-optical and by molecular CN in the near-infrared. NH, the other molecule Asplund et al. 2021 use, has no usable band in our spectra. The top number is the plain mean of the atomic and molecular results, as theirs is. An unresolved or unconstrained fit supplies no adopted Solar nitrogen abundance.

C/O indicator strategy · Methodology · Nitrogen strategy

Open provenance decisions

NIST's N I values ARE Tachiev (TP T7370). Adjudicating N I against this pull is one source seen twice, not independent confirmation.

Adjudication record · Indicator gf checks

Download the evidence

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Source feed — includes quarantined fit outputs, not adopted abundances

Complete product visibility audit

Shared website/PDF report

Reproducibility manifest

Complete references and product-to-source bindings

Reproducibility

Generated from the pinned science snapshot. Source commit 46fd923d510159bad5c0aaf5eeb616e175e048e5
Feed N.json v1.80 · updated 2026-10-10T20:48:37Z
Generated 2026-10-29T07:34:56-06:00 (source commit timestamp)
Feed SHA-256 e0c02e52a4ef9818865ae7543d9a08169ff453580c49b3c6dacad3ce303a747e
Generator scripts/generate_cno_publication.py v1.0.0

Pinned scientific source

References / Data & Model Sources

Sources are separated by their role. Registered grid lineage and quarantined diagnostics do not imply an adopted abundance or an applied correction.

See the source notes for the distinction between reference lineage and the treatment actually recorded in each feed entry.

Science and provenance source notes

Solar abundance and indicator analyses. Solar N atomic/molecular indicator lineage and atmospheric sensitivity; comparison literature, not an adopted Codex abundance. Sources: [1], [2].

Atomic data and gf authorities. The recorded N I/II pedigree is the partial NIST MCHF update described by Wiese & Fuhr; evaluated theory is not primary laboratory evidence. Sources: [3], [4].

Atomic data and gf authorities. NIST's N I values ARE Tachiev (TP T7370). Adjudicating N I against this pull is one source seen twice, not independent confirmation. Sources: [3], [5].

Model and grid sources. The N I registry uses the Amarsi GALAH/PySME 1D-NLTE departure grid. The solar 3D-NLTE atomic analysis is a separate source, not a full-3D correction applied by this feed. Sources: [6], [7], [8].

Molecular line data and delivery. CN line-data delivery: CN red A-X: Brooke et al. 2014 / Sneden et al. 2014 (isotopologues). Masseron compilation, VALD-based, distributed with iSpec/Turbospectrum (Gerber et al. 2023, A&A 669, A43; uploaded 2022 to keeper.mpdl.mpg.de, split per wavelength region) — see molecules-dir README.md Sources: [2], [9], [10], [11].

Model and grid sources. Feed treatment: 1D-LTE; no NLTE/3D correction is claimed for this entry. Turbospectrum is the synthesis implementation. Sources: [9].

Atomic data and gf authorities. Atomic N I entry; source-family pedigree only, not a substitute for a per-transition gf join. Sources: [3], [4], [5].

Model and grid sources. Feed label: Synth · 3D-NLTE · Amarsi. Cited per-line solar correction, Amarsi et al. 2020 Table 3 (3D non-LTE minus 1D LTE); not an in-transfer 3D-NLTE result. Run provenance: Amarsi, Grevesse, Grumer, Asplund, Barklem & Collet 2020, A&A 636, A120, Table 3 (solar, disk-centre EWs) -- 3D non-LTE (STAGGER) - 1D LTE (MARCS); SOLAR per-line additive correction, same EWs in both Sources: [6], [7], [8].

Model and grid sources. Feed label: Synth · 1D-NLTE · Bergemann. Registered source: Amarsi/PySME N I 1D-NLTE. The generic Bergemann label is retained as supplied and is not used as a bibliographic attribution. Sources: [6], [7], [8].

Product-to-source index

NIR · CRIRES+ J · CN A–X (0–0) band, near-infrared (molecular band) · 1D LTE · [1], [2], [9], [10], [11] N-products-f80b21113bb7

NIR · Kitt Peak (1984, molecfit) · CN A–X (0–0) band, near-infrared (molecular band) · 1D LTE · [1], [2], [9], [10], [11] N-products-bbb7416bb60c

VIS · HARPS · CN red system, visible (molecular band; diagnostic only) · 1D LTE · [1], [2], [9], [10], [11] N-products-04b24d6c2d67

red-optical · IAG FTS · N I 8629 + 8683 Å, the two least-blended lines (Magg et al. 2022, adopted by Lodders et al. 2025) · 1D LTE · [1], [3], [4], [5], [9] N-products-854e3ddc73f8

red-optical · IAG FTS · N I 8629 + 8683 Å, the two least-blended lines (Magg et al. 2022, adopted by Lodders et al. 2025) · 3D NLTE (Amarsi) · [1], [3], [4], [5], [6], [7], [8] N-products-95a9d8d61cb1

red-optical · IAG FTS · N I 8629 + 8683 Å, the two least-blended lines (Magg et al. 2022, adopted by Lodders et al. 2025) · 1D NLTE (Bergemann) · [1], [3], [4], [5], [6], [7], [8] N-products-74b4db6b5e6a

red-optical · Kitt Peak (Kurucz 2005) · N I 8629 + 8683 Å, the two least-blended lines (Magg et al. 2022, adopted by Lodders et al. 2025) · 1D LTE · [1], [3], [4], [5], [9] N-products-9ad55347df16

red-optical · Kitt Peak (Kurucz 2005) · N I 8629 + 8683 Å, the two least-blended lines (Magg et al. 2022, adopted by Lodders et al. 2025) · 3D NLTE (Amarsi) · [1], [3], [4], [5], [6], [7], [8] N-products-3feb158ef160

red-optical · Kitt Peak (Kurucz 2005) · N I 8629 + 8683 Å, the two least-blended lines (Magg et al. 2022, adopted by Lodders et al. 2025) · 1D NLTE (Bergemann) · [1], [3], [4], [5], [6], [7], [8] N-products-c22e5b580ebb

red-optical · Kitt Peak (1984, molecfit) · N I 8629 + 8683 Å, the two least-blended lines (Magg et al. 2022, adopted by Lodders et al. 2025) · 1D LTE · [1], [3], [4], [5], [9] N-products-db4e37f1c3b3

red-optical · Kitt Peak (1984, molecfit) · N I 8629 + 8683 Å, the two least-blended lines (Magg et al. 2022, adopted by Lodders et al. 2025) · 3D NLTE (Amarsi) · [1], [3], [4], [5], [6], [7], [8] N-products-8abe06adf8d9

red-optical · Kitt Peak (1984, molecfit) · N I 8629 + 8683 Å, the two least-blended lines (Magg et al. 2022, adopted by Lodders et al. 2025) · 1D NLTE (Bergemann) · [1], [3], [4], [5], [6], [7], [8] N-products-ed75c72efb5b

red-optical · IAG FTS · N I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [3], [4], [5], [6], [7], [8] N-quarantine-fc1525ccd677

red-optical · Kitt Peak (Kurucz 2005) · N I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [3], [4], [5], [6], [7], [8] N-quarantine-9cb72bc628d8

red-optical · Kitt Peak (1984, molecfit) · N I · Asplund et al. 2021 lines · 1D LTE · [1], [3], [4], [5], [9] N-quarantine-35945ad2b73d

red-optical · Kitt Peak (1984, molecfit) · N I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [3], [4], [5], [6], [7], [8] N-quarantine-4d3667606a8f

red-optical · Kitt Peak (1984, molecfit) · N I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [3], [4], [5], [6], [7], [8] N-quarantine-66d14a326159

Complete applicable bibliography

Solar abundance and indicator analyses

[1] — Asplund, M.; Amarsi, A. M.; Grevesse, N. (2021). The chemical make-up of the Sun: A 2020 vision. A&A 653, A141 DOI: 10.1051/0004-6361/202140445. source record

[2] — Amarsi et al. 2021, A&A 656 A113 DOI: 10.1051/0004-6361/202141384. source record

Atomic data and gf authorities

[3] — Kramida, A.; Ralchenko, Yu.; Reader, J.; NIST ASD Team (2024). NIST Atomic Spectra Database (version 5.11/5.12). National Institute of Standards and Technology, Gaithersburg, MD DOI: 10.18434/T4W30F. source record

[4] — Wiese, W. L.; Fuhr, J. R. (2006). New Critical Compilations of Atomic Transition Probabilities for Neutral and Singly Ionized Carbon, Nitrogen, and Iron. Proceedings of the NASA Laboratory Astrophysics Workshop (14-16 February 2006, UNLV, Las Vegas) source record

[5] — Tachiev & Froese Fischer 2002, A&A 385, 716 DOI: 10.1051/0004-6361:20011816. source record

Model and grid sources

[6] — Amarsi et al. 2020, A&A 642 A62 DOI: 10.1051/0004-6361/202038650. source record

[7] — Amarsi, Grevesse, Grumer, Asplund, Barklem & Collet 2020, A&A 636, A120 DOI: 10.1051/0004-6361/202037890. source record

[8] — Amarsi 'Grid/NLTE' Zenodo, version 3 (the original 13-element 2020 release) DOI: 10.5281/zenodo.3982506. source record

Molecular line data and delivery

[9] — Gerber, J. M.; Magg, E.; Plez, B.; Bergemann, M.; Heiter, U.; Olander, T.; Hoppe, R. (2023). Non-LTE radiative transfer with Turbospectrum. A&A 669, A43 DOI: 10.1051/0004-6361/202243673. source record

[10] — Brooke et al. 2014, ApJS 210, 23 DOI: 10.1088/0067-0049/210/2/23. source record

[11] — Sneden, C.; Lucatello, S.; Ram, R. S.; Brooke, J. S. A.; Bernath, P. (2014). Line Lists for the Red and Violet Systems of CN, 13C14N, and 12C15N, and Application to Astronomical Spectra. ApJS 214, 26. DOI: 10.1088/0067-0049/214/2/26. source record